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The Solar Generator Buyer's Guide 2026

How to size, compare, and safely use a portable power station for outages, camping, RVs, and home backup — in plain English.

- The sizing math, with a fill-in worksheet
- The specs that actually matter (and the ones that don't)
- Solar panel recharge math and cold-weather rules
- Safety basics, printable outage checklists, and a buyer's checklist

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How to Use This Guide

Most people shop for a solar generator backwards: they compare brands first and work out what they need later. This guide flips that. Work through it in order and you will know the size, features, and budget you need *before* you look at a single product page.

1. **Step 1 — List what you need to power**: typical wattages and a fill-in worksheet.
2. **Step 2 — Do the sizing math**: turn your list into watt-hours and a target capacity.
3. **Step 3 — Read the specs that matter**: capacity, output, surge, battery chemistry, and more.
4. **Step 4 — Plan your solar**: realistic panel output and recharge times.
5. **Step 5 — Cold, storage, and safety**: the rules that protect you and the battery.
6. **Step 6 — Match a size to your situation**: what each budget tier really gets you.
7. **Checklists**: blackout cheat sheet, storm timeline, and buyer's checklist.

> **How we research.** S.E.T. compares products using manufacturer spec sheets cross-checked against independent third-party test data and owner reports, then judges them by use case. We do not run our own lab tests, so every runtime in this guide is a clearly labeled *estimate* with the math shown. Specs and prices change often — always confirm on the manufacturer's current spec page before you buy.

Three numbers you will see everywhere

Term	What it means	Analogy
Watts (W)	How much power a device draws *right now*, or how much a power station can deliver at once.	Speed of water through a hose
Watt-hours (Wh)	How much energy is stored or used over time. 100W for 10 hours = 1,000Wh (1kWh).	Size of the water tank
Surge watts	A short burst of extra power for motor startups (fridges, pumps, AC compressors).	A quick extra push to get the wheel turning

Tip: if a spec sheet lists capacity only in amp-hours (Ah), multiply by the battery voltage to get Wh. 100Ah at 12.8V ≈ 1,280Wh.

STEP 1

List What You Need to Power

The fastest way to get real numbers: check the label on the device or its charger, or plug it into an inexpensive plug-in watt meter for a day. For refrigerators, the yellow EnergyGuide label lists **kWh per year** — divide by 365 for kWh per day.

Device	Typical running watts	Typical daily use	Startup surge?
Full-size refrigerator (newer)	100–200W while the compressor runs	About 1–2 kWh/day	Yes, 3–5× for a split second
Chest freezer	60–150W while running	About 0.5–1.5 kWh/day	Yes
CPAP (no heated humidifier)	30–60W	0.25–0.5 kWh/night	No
CPAP with heated humidifier	Can double or triple the draw	Check your model	No
Wi-Fi router + modem	10–25W	0.25–0.6 kWh/day	No
LED bulb	8–12W each	Hours used × watts	No
Phone (full charge)	—	About 10–20Wh per charge	No
Laptop	45–100W while charging	Hours used × watts	No
Box or ceiling fan	30–100W	Hours used × watts	Small
Microwave	1,000–1,500W	Short bursts	Small
Window AC	500–1,200W	Large — plan carefully	Yes
Space heater	Up to 1,500W	1,500W × 8h = 12 kWh (very large)	No
Sump pump	800–1,500W	Depends on rain	Yes, 2–3×

Usually not practical on a portable unit: electric water heaters, electric dryers, electric ranges, central AC, and most 240V well pumps. These need a whole-home battery system or a properly installed generator.

Your worksheet

Device	Watts	Hours per day	Wh per day (W × h)
Total	*Peak W (things on at once):*		*Daily Wh:*

Typical ranges are general estimates; your devices may differ. Measure when you can.

STEP 2

Do the Sizing Math

You need two numbers from your worksheet: **daily watt-hours** (how big the battery must be) and **peak watts** (how strong the inverter must be).

1. Capacity: how many watt-hours?

Not all of the rated capacity reaches your devices. Inverter and conversion losses mean you should plan on roughly **85% usable**.

```
Rated capacity needed ≈ Daily Wh ÷ 0.85 × 1.2 (20% safety margin)
```

2. Output: how many watts?

Add up everything that could run *at the same time*. The continuous output rating must exceed that number, and the surge rating must cover the biggest motor starting up.

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Continuous output > peak running watts   |   Surge rating > peak watts + largest startup surge
```

Worked example: a one-day outage for a family

Device	Estimate	Wh per day
Newer full-size fridge (EnergyGuide 548 kWh/yr)	548 ÷ 365	1,500
Wi-Fi router + modem	15W × 24h	360
CPAP (no humidifier)	40W × 8h	320
4 LED bulbs	4 × 10W × 5h	200
4 phones	4 × 15Wh	60
Total		**2,440 Wh**

```
2,440 ÷ 0.85 × 1.2 ≈ 3,445 Wh rated capacity for a full day with margin
```

That points to a **3kWh-class** unit for a full 24 hours without recharging — or a **2kWh-class** unit plus solar panels to top it up during the day (see Step 4). Peak watts here are low (fridge + router + lights ≈ 250W), so almost any 2kWh-class inverter handles the load; the fridge's startup surge is the number to check.

> **Shortcut:** turn the router off overnight and charge phones in the car; trimming non-essentials can shrink your daily number, and the price of the unit with it.

How long will a given unit last? (estimate)

```
Runtime (hours) ≈ Rated Wh × 0.85 ÷ average watts
e.g. 2,048Wh × 0.85 ÷ 40W CPAP ≈ 43 hours (estimate; varies with temperature and load)
```

STEP 3

The Specs That Actually Matter

1. Capacity (Wh) — how long it runs

The single most important number. Compare usable energy, not marketing claims: plan on about 85% of the rated Wh reaching your devices.

2. Continuous output (W) — what it can run

The maximum steady load. Make sure it exceeds everything you will run at once. Look for a **pure sine wave** inverter; it is safe for CPAPs, laptops, and motors.

3. Surge output (W) — motor startups

Refrigerators, pumps, and AC compressors pull 2–5× their running watts for a split second. Too little surge and the unit trips its overload protection.

4. Battery chemistry — how many years it lasts

Chemistry	Typical rated cycle life	Pros	Cons
LiFePO4 (LFP)	About 3,000–6,000 cycles to 70–80% capacity	Long life, more thermally stable, most new models	Heavier per Wh; can't be charged below freezing without heating
NMC / Li-ion	Often 500–1,000 cycles to 80%	Lighter for its capacity	Shorter life; older or budget models

One full cycle a day for 3,000 cycles is more than 8 years. Check what capacity the rating is measured to (70% vs 80%) when you compare.

5. Solar input (W and voltage range) — how fast it refills

Higher solar input means faster recharges. Also check the input **voltage range**: your panels' combined open-circuit voltage (Voc) must stay below the maximum, especially on cold mornings when Voc rises.

6. UPS / pass-through switchover — for sensitive gear

Some units switch to battery automatically when the grid drops. A switchover of roughly 30 milliseconds or less usually keeps routers, desktops, and CPAPs from rebooting. Check the spec sheet; not every unit offers it.

Spec Translator: The Rest of the Sheet

Spec	What it means for you
AC charging rate (W)	How fast it refills from a wall outlet. Higher means more charged before a storm hits.
AC + solar dual charging	Charges from the wall and panels at the same time for the fastest refill.
Expandability	Add-on batteries increase capacity later. Check the maximum and the cost per extra kWh.
MPPT controller	The built-in solar charger. MPPT gets more energy from the same panels than older PWM controllers.
Outlets	Count the AC outlets you need. RVs usually need a 30A (TT-30) outlet or a properly rated adapter.
USB-C PD (W)	Charges laptops directly without the less efficient AC inverter. 100W+ ports are handy.
12V car port	Runs 12V fridges and CPAP DC cords efficiently.
Weight and wheels	Under ~30 lbs is a one-hand carry. 50–65 lbs needs two hands, a cart, or built-in wheels.
Noise	Fans run harder during fast charging and heavy loads. Many units let you slow the charge rate overnight.
Operating and charging temperature	Two separate ranges. Charging range is usually narrower — important in cold climates.
Warranty and support	3–5 years is common on LFP units. Check who handles repairs and where.
Certifications	Look for recognized safety listings (for example UL or ETL marks) on the product or spec sheet.

“X-Boost” and similar modes let some units run resistive devices (kettles, heaters) rated above the inverter’s limit by lowering the voltage. The device runs at reduced power. It does not help motors start.

Watch for: capacity listed only in mAh, “peak” watts shown larger than continuous watts, or no stated cycle-life rating. These usually signal a weaker product.

STEP 4

Plan Your Solar Recharging

Solar panels almost never deliver their full rated watts. Heat, angle, haze, and cable losses mean portable panels typically deliver about **70–85% of their rating** in good sun. That is physics, not a defect.

```
Hours of strong sun to refill ≈ Usable Wh ÷ (panel watts × 0.8)
```

Example	Math	Estimate
2,048Wh unit, 400W of panels	1,740 ÷ (400 × 0.8)	≈ 5.4 hours of strong sun
2,048Wh unit, 800W of panels	1,740 ÷ (800 × 0.8)	≈ 2.7 hours
1,024Wh unit, 200W of panels	870 ÷ (200 × 0.8)	≈ 5.4 hours

Peak sun hours

A "peak sun hour" is one hour of full-strength sun. Across the U.S., most places get roughly **3–6 peak sun hours a day**, and much less in winter. Size your panels for the season you care about most.

Getting the most from your panels

- **Avoid shade.** Shading part of one panel can cut the output of the whole string.
- **Aim them.** Face the sun and re-angle a few times a day; steeper tilts help in winter and shed snow.
- **Stay inside the input limits.** Keep the total Voc of panels wired in series below the unit's maximum input voltage, with margin for cold mornings.
- **Keep them clean.** Dust, pollen, and bird droppings measurably reduce output.
- **Mind the cable.** Use the shortest, thickest extension cable you can.

Portable panels vs. permanent

Portable (folding) panels are easy to store and aim, ideal for camping and short outages. Setup each time is the trade-off.

Fixed or ground-mounted panels deliver more watts per dollar and need no daily setup — better for frequent outages or off-grid use.

> **Reality check:** if your daily use is 2,400Wh and you get 4.5 peak sun hours, you need roughly 2,400 ÷ (4.5 × 0.8) ≈ **670W of panels** to replace a full day of use from solar alone.

STEP 5

Cold Weather, Storage, and Safety

Cold weather

- Most LiFePO4 batteries must **not be charged below 32°F (0°C)** unless the unit has self-heating or a low-temperature charge cutoff. Check the charging temperature range.
- Expect noticeably less usable capacity in freezing temperatures. Keep the unit indoors or insulated and warm it before charging.

Storage between outages

- Follow the manual. Many manufacturers recommend storing at a partial charge and topping up every few months.
- Keep it cool and dry, out of direct sun, and off damp floors.
- Run a quick test every couple of months: charge it fully, plug in your most important device for a few minutes, and confirm everything works.

Safety rules that matter most

- **Never run a gas generator indoors, in a garage, or near open windows.** Carbon monoxide kills. Keep battery-powered CO alarms working during outages.
- **Never plug a generator or power station into a wall outlet** to feed the house ("backfeeding"). It can electrocute utility workers and damage equipment. Use extension cords, or have an electrician install a transfer switch or interlock.
- **Don't cover the vents** while charging or running heavy loads, and keep the unit away from water.
- **Use properly rated cords.** Heavy loads need heavy-gauge extension cords; warm cords are a warning sign.
- **Medical devices:** talk to your provider about backup plans, and register with your utility's medical-priority program if one is offered.

Solar generator vs. gas generator

	Battery power station	Gas generator
Indoor use	Yes, no fumes	Never
Noise	Quiet; fans only	Loud
Fuel	Sun or wall outlet	Gasoline storage and refueling
Multi-day heavy loads	Limited by capacity and sun	Strong, as long as fuel lasts
Maintenance	Minimal	Oil, carburetor, fuel stabilizer
Best use	Quiet indoor essentials	Long outages, heavy loads (outdoors only)

STEP 6

Match a Size to Your Situation

Price ranges below are typical for LFP portable power stations in late 2026 and change often with sales. Use them to set expectations, then compare specific models on our site.

Typical price	Capacity class	What it realistically covers
Under ~$300	250–500Wh	Phones, lights, a router, one night of CPAP. Great for camping and short outages.
~$300–$800	500–1,100Wh	A CPAP for several nights, laptops, fans, or a fridge for part of a day.
~$800–$1,500	1–2kWh	A newer fridge for roughly a day plus small essentials. The sweet spot for many homes.
~$1,500–$3,000+	2–5kWh, often expandable	Multi-day essentials with solar, small window AC, RV use, bigger combined loads.
Installed system	10kWh+ home battery	Whole-home or critical-circuit backup through a transfer device. A different category with professional installation.

By situation

Situation	Prioritize
Power outages at home	Capacity for your daily Wh, enough surge for the fridge, UPS switchover for routers and CPAPs, fast AC charging before storms.
CPAP and medical	Reliable pure-sine output, UPS switchover, enough Wh for 2–3 nights, a DC cord option for efficiency.
Camping and tailgating	Weight, solar input, USB-C ports, quiet operation.
RV and van life	2kWh+, high solar input, a 30A RV outlet or adapter, vehicle charging, space under a bench.
Hurricane and storm regions	Multi-day capacity, solar plus expandability, fast pre-storm charging.
Cold climates	Low-temperature charging protection or self-heating, a place to keep it warm.
Apartments	Fits indoors, quiet, no fumes; check building rules on battery storage.

> **Compare current models:** our side-by-side table (sunenergytechnology.com/compare.html) lets you filter by budget, what you're powering, and use case, with runtime estimates for each model.

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Blackout Cheat Sheet

Before the outage

- [ ] Charge the power station to 100%
- [ ] Test it with the fridge and router for a few minutes
- [ ] Charge phones, laptops, and power banks
- [ ] Set out extension cords, a flashlight, and the manual
- [ ] Check CO and smoke alarm batteries
- [ ] Turn the fridge and freezer to their coldest settings
- [ ] Know where your main breaker is

During the outage

- [ ] Plug in essentials only: fridge, router, CPAP, lights
- [ ] Keep fridge and freezer doors closed
- [ ] Turn off devices you're not using
- [ ] Recharge from solar in the middle of the day
- [ ] Check the battery percentage morning and evening
- [ ] Never backfeed a wall outlet

After power returns

- [ ] Recharge the station fully before putting it away
- [ ] Check food safety: a full freezer holds about 48 hours closed, a fridge about 4 hours without power
- [ ] Note what ran out first and adjust your plan

My essentials and runtime estimates

Device	Average watts	Est. hours (unit Wh × 0.85 ÷ W)

Food safety guidance from the USDA: keep doors closed; discard perishable food held above 40°F for more than 2 hours.

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Storm Timeline & Buyer's Checklist

Storm-season timeline

When	Do this
Start of season	Test the station and panels, update firmware, replace worn cords, review this guide's worksheet.
72 hours out	Charge everything to 100%. Stock water, food, and medications. Confirm your CO alarms work.
24 hours out	Top up again. Freeze water bottles to help keep the fridge cold. Bring panels and the station indoors.
During the storm	Stay inside. Run essentials only. Keep panels stored until winds pass.
After the storm	Set panels up only when safe. Watch for downed lines. Recharge midday; ration evening use.

Buyer's checklist

- [ ] My daily Wh need is _____ and my peak watts are _____ (Steps 1–2)
- [ ] Rated capacity covers my daily Wh ÷ 0.85 with margin
- [ ] Continuous output exceeds my peak watts; surge covers my fridge or pump
- [ ] Pure sine wave inverter
- [ ] LiFePO4 (LFP) battery with a stated cycle-life rating
- [ ] Solar input and voltage range fit the panels I plan to use
- [ ] UPS / pass-through switchover if I run a router, PC, or CPAP
- [ ] Charging temperature range works for where I'll keep it
- [ ] Enough of the right outlets (AC, USB-C, 12V, RV 30A)
- [ ] Weight I can actually move, or wheels
- [ ] Expandable if I may need more capacity later
- [ ] Warranty of 3+ years and a clear repair process
- [ ] Recognized safety certification
- [ ] Price checked against the current sale price, not the list price

Three mistakes buyers make

1. **Buying on capacity alone** and finding the inverter can't start the fridge.
2. **Assuming rated panel watts** and expecting a full recharge on a cloudy day.
3. **Leaving it in the garage all winter** and discovering it won't charge in the cold.

Where to Go Next

- **Compare models side by side:** sunenergytechnology.com/compare.html — filter by budget, what you're powering, and use case, with runtime estimates.
- **Size your own loads:** sunenergytechnology.com/calculator.html
- **Decode a spec sheet:** sunenergytechnology.com/spec-guide.html
- **Read reviews and head-to-heads:** sunenergytechnology.com/blog/
- **Thinking about home batteries or rooftop solar?** sunenergytechnology.com/home-batteries.html

You're also enrolled in the free S.E.T. Solar Buyer's Course: short, plain-English emails over the next three weeks that go deeper on battery life, solar panels, home backup, and storm prep. Reply to any of them with questions.

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